

Nudt2 (P56380, Mouse): Functional Annotation Report

Gene: *Nudt2* (synonym *Apah1*) | **UniProt:** P56380 | **Organism:** *Mus musculus* (mouse) **Protein:** Bis(5'-nucleosyl)-tetraphosphatase [asymmetrical] / Diadenosine tetraphosphate (Ap4A) hydrolase | **EC 3.6.1.17** | **Family:** Nudix hydrolase (Nudix motif 2)

Summary

Mouse **Nudt2** encodes a small (~17 kDa, 147 aa), soluble, predominantly **nuclear** enzyme of the **Nudix hydrolase** superfamily. Its primary, defining activity is the **metal-dependent asymmetrical hydrolysis of diadenosine 5',5'''-P¹,P⁴-tetraphosphate (Ap4A)**, in which the polyphosphate chain is cleaved at the fourth phosphate from one adenosine, yielding **ATP + AMP** (EC 3.6.1.17). The enzyme acts on a broader class of dinucleoside 5',5'''-polyphosphates (Ap5A, Ap6A, Gp4G, Gp5G), always releasing a **nucleoside triphosphate plus the remaining nucleotide moiety**, and requires a divalent cation (Mg²⁺ or Zn²⁺) coordinated by conserved glutamates in the Nudix catalytic box. Verification against the UniProt record, domain architecture, and cross-species sequence conservation confirms that the gene symbol "Nudt2" correctly matches this protein; the mouse protein is 89% identical to human NUDT2 and retains an invariant catalytic Nudix loop.

Biologically, Nudt2 is the **homeostatic "off-switch" for the Ap4A alarmone**, a stress-signaling dinucleotide that accumulates during heat and oxidative stress and that functions as an intracellular messenger. Ap4A is produced as a side-product by lysyl-tRNA synthetase (LysRS) during immune activation and acts within the **LysRS → Ap4A → Hint1 → MTF** transcriptional signaling axis. By degrading Ap4A, Nudt2 controls the amplitude and duration of this signal. Experimental disruption of NUDT2 causes a **175-fold rise in intracellular Ap4A** and thousands of gene-expression changes, down-regulating interferon/inflammation pathways and up-regulating MHC class II genes. A second, mechanistically distinct catalytic activity has emerged: NUDT2 is a bona fide **mRNA decapping enzyme** that removes non-canonical Ap4A 5' RNA caps, thereby regulating mRNA half-life and stability.

The clinical and physiological importance of this enzyme is underscored by the recent discovery that **biallelic loss-of-function variants in human NUDT2 cause a recessive neurodevelopmental disorder**, and that the mRNA-decapping activity specifically — not Ap4A hydrolysis alone — is responsible for the altered mRNA homeostasis in patient cells. Because the mouse enzyme is highly conserved and shares the complete catalytic machinery, these human and cross-species data are directly transferable to the annotation of mouse Nudt2. This report synthesizes ten confirmed findings across enzymology, structural biology, cell signaling, subcellular localization, evolution, and disease genetics into a coherent functional model.

Gene/Protein Identity Verification

Before proceeding, the mandatory identity checks were satisfied:

Check	Result
Gene symbol matches protein description	☐ "Nudt2" = Nudix motif 2 = Ap4A hydrolase, consistent with UniProt P56380
Organism correct	☐ <i>Mus musculus</i>
Protein family / domains align with literature	☐ Nudix hydrolase fold; conserved Nudix box verified in sequence (Finding F009)
No confusion with same-symbol gene in another organism	☐ Human (P50583), pig (P50584), <i>Drosophila</i> and bacterial orthologs all encode the same asymmetrical Ap4A hydrolase activity (Findings F001, F002, F010)

The literature for this enzyme class is **consistent and unambiguous**: across mouse, human, pig, *Drosophila*, and bacteria, the "Nudt2/Apah1/Ap4A hydrolase/ApaH/IalA" gene products all catalyze asymmetrical hydrolysis of dinucleoside polyphosphates. No conflicting same-symbol gene was found.

Key Findings

F001 — Nudt2 is an asymmetrical Ap4A hydrolase (EC 3.6.1.17) of the Nudix family

UniProt P56380 annotates mouse Nudt2 (Apah1) as **bis(5'-nucleosyl)-tetraphosphatase, asymmetrical** — the Ap4A hydrolase, EC 3.6.1.17 — a member of the Nudix hydrolase family bearing Nudix motif 2. The defining feature of the *asymmetrical* enzyme is its regiospecificity. The crystal structure of the human ortholog shows the enzyme "**cleaving the polyphosphate chain at the fourth phosphate from the bound adenosine moiety**" and adopting "**the common $\alpha\beta\alpha$ -sandwich architecture**," the canonical Nudix fold (PMID: 23384440). This geometry explains why the products are **ATP + AMP** rather than two ADP molecules (the latter being the product of the unrelated *symmetrical* Ap4A hydrolases). The asymmetrical cleavage is the signature that distinguishes the Nudt2/Nudix enzyme from symmetrical Ap4A hydrolases of other families.

Reaction: $\text{Ap4A} + \text{H}_2\text{O} \rightarrow \text{ATP} + \text{AMP}$ (asymmetrical cleavage of the polyphosphate chain at the fourth phosphate relative to one adenosine)

F002 — Substrate specificity: dinucleoside polyphosphates cleaved to NTP + NMP, divalent-cation dependent

The enzyme's substrate preference and metal dependence are best quantified in orthologs. The *Drosophila* asymmetrical Ap4A hydrolase hydrolyzes Ap4A to **ATP + AMP** with **$K_m \approx 9 \mu\text{M}$ and $k_{cat} \approx 43 \text{ s}^{-1}$** (pH 6.5, 0.1 mM Zn^{2+}); it also processes Ap5A and Ap6A, with the product and catalytic efficiency depending on pH and the divalent ion present (Zn^{2+} or Mg^{2+}). Fluoride inhibits the Mg^{2+} -dependent reaction with an IC_{50} of $\sim 20 \mu\text{M}$, acting through formation of an **MgF_3^- transition-state analogue**, which provides mechanistic evidence for a metal-assisted phosphoryl-transfer chemistry (PMID: 17344088). Quoting the kinetic study directly: "**diadenosine tetraphosphate is hydrolysed to ATP and AMP.**"

The breadth of the dinucleoside polyphosphate specificity is illustrated by the *Bartonella* Nudix ortholog IalA, which shows "**highest activity on adenosine 5'-tetraphospho-5'-adenosine (Ap4A), but also hydrolyzing Ap5A, Ap6A, Gp4G, and Gp5G ... a pyrophosphate linkage is**

cleaved yielding a nucleoside triphosphate and the remaining nucleotide moiety" (PMID: 9880487). Thus the enzyme is **Ap4A-preferring but Ap_nA/Gp_nG-general**, always releasing an NTP + NMP.

Property	Value / substrate	Source
Preferred substrate	Ap4A	P 9880487
		P 17344088
Additional substrates	Ap5A, Ap6A, Gp4G, Gp5G	P 9880487
Products	NTP + NMP (Ap4A → ATP + AMP)	P 17344088
		P 9880487
Km (Ap4A, <i>Drosophila</i>)	~9 μM	P 17344088
kcat (Ap4A, <i>Drosophila</i>)	~43 s ⁻¹	P 17344088
Metal requirement	Mg ²⁺ or Zn ²⁺	P 17344088
Inhibitor	Fluoride (IC ₅₀ ~20 μM; MgF ₃ ⁻ TS analogue)	P 17344088

F003 — Nudt2 controls intracellular Ap4A levels and thereby immune/cancer gene expression

The causal, in-cell role of NUDT2 as the master regulator of Ap4A was established by gene disruption. In KBM-7 cells, disruption of **"the NUDT2 Ap4A hydrolase gene ... (NuKO cells), causing a 175-fold increase in intracellular Ap4A"** produced **6,288 differentially expressed genes** ($P < 0.05$) (PMID: 27144453). Interferon, pattern-recognition, and inflammation pathways were **down-regulated** while **MHC class II genes were up-regulated**, with predicted upstream regulators including NF-κB, STAT1/2, IRF3/4, and SP1. This demonstrates that NUDT2 is the **dominant determinant of intracellular Ap4A concentration** and that Ap4A homeostasis is transcriptionally consequential — connecting the enzyme to immune and cancer-relevant gene programs.

F004 — Ap4A/Nudt2 operates in the LysRS–Hint1–MITF signaling axis

Nudt2's substrate, Ap4A, is a genuine intracellular second messenger. In FcεRI-activated mast cells, **lysyl-tRNA synthetase (LysRS)** synthesizes Ap4A (as a side-product of the condensation of Lys-AMP with ATP), which accumulates above **700 μM**, binds the repressor **Hint1**, and **"binds to Hint, liberates MITF, and thus leads to the activation of MITF-dependent gene expression"** (PMID: 14975237). The upstream regulation of Ap4A production is itself tightly controlled: **"LysRS was phosphorylated on serine 207 in a MAPK-dependent manner, released from the multisynthetase complex, and translocated into the nucleus"** during immune challenge (PMID: 19524539). In this pathway, **LysRS is the signal generator and Nudt2 is the signal terminator** — the enzyme that degrades Ap4A back to ATP + AMP and resets the system. The structural basis of Hint1's recognition of aminoacyl adenylates, which constrains Ap4A production, is further detailed in PMID: 22329685.

F005 — Subcellular localization: predominantly nuclear

The asymmetrical Ap4A hydrolase localizes to the nucleus. An Apf-EGFP fusion of the *Drosophila* ortholog **"reveals Apf to be predominantly nuclear, having an apparent preferential association with euchromatin and facultative heterochromatin,"** consistent with a nuclear function for Ap4A metabolism (PMID: 17344088). The mouse protein is a small soluble ~16–17 kDa protein with no transmembrane segments (Finding F009), compatible with nucleocytoplasmic distribution. Importantly, the **extracellular** degradation of Ap4A is carried out by a *different* enzyme: **"the activity previously described as bovine adrenal chromaffin cell ecto(diadenosine polyphosphate hydrolase) (ecto-ApnAase) is a PDase-I"** (alkaline phosphodiesterase-I / PC-1), not the intracellular Nudix Ap4A hydrolase (PMID: 9784621). This cleanly separates Nudt2's **intracellular/nuclear** role from the ecto-nucleotidase pathways that act on extracellular dinucleotides.

F006 — Ap4A is a non-canonical 5' RNA cap, implicating Nudt2 in RNA cap metabolism

Beyond its role as a free metabolite, Ap4A can serve as a **5' RNA cap**. A 2024 study reports **"the existence of a new non-canonical RNA cap, diadenosine tetraphosphate (Ap4A)"** in eukaryotes and notes that **"the discovery of non-canonical RNA caps in eukaryotes revealed a new niche of previously undetected RNA chemical modifications"** (PMID: 37934413). Because Ap4A is Nudt2's canonical substrate, this positions the enzyme as a candidate **decapping** enzyme for Ap4A-capped transcripts — a prediction borne out directly by the disease genetics (F008).

F007 — Ap4A is a conserved stress "alarmone"; its hydrolase protects against stress

Ap4A (AppppA) **"is rapidly synthesized in cells exposed to heat stress or oxidative stress,"** and **"stress-induced AppppA accumulation has been observed in all cell types studied to date."** Critically, bacterial *apaH* (Ap4A hydrolase) mutants, which cannot degrade Ap4A, accumulate it and **"apaH mutants, which have high basal levels of AppppA, are hypersensitive to killing by heat"**; Ap4A also binds stress proteins DnaK and GroEL ([PMID: 1935909](#)). This is the clearest evidence that **loss of the Ap4A hydrolase is deleterious under stress** — the hydrolase has a protective, homeostatic role. The alarmone concept was proposed earlier: these adenylylated dinucleotides **"may be alarmones—i.e., regulatory molecules, alerting cells to the onset of oxidation stress"** ([PMID: 6369319](#)). ApaH is the bacterial functional counterpart of Nudt2, so these results predict that Nudt2 loss should likewise elevate Ap4A and compromise stress responses.

F008 — NUDT2 is a bona fide mRNA decapping enzyme; loss causes a human neurodevelopmental disorder

This is the most important recent advance in understanding the enzyme. **Biallelic rare variants in NUDT2 cause a recessive neurological disorder** in 18 children/young adults from 10 families, with intellectual disability, motor delay, gait disturbance, and sometimes peripheral neuropathy and basal-ganglia deposits. NUDT2 is described as **"a mRNA decapping and Ap4A hydrolysing enzyme,"** and disease variants **"lead to a marked loss of enzymatic activity."** NUDT2-deficient fibroblasts show an altered transcriptome with changes in **"mRNA half-life and stability,"** up-regulating **"host response and interferon-responsive genes."** Decisively, **"add-back experiments using an Ap4A hydrolase defective in mRNA decapping highlighted loss of NUDT2 decapping as the activity implicated in altered mRNA homeostasis"** ([PMID: 38141063](#)). This genetically separates the two activities and shows that it is the **decapping** function that governs mRNA stability.

An independent report confirms that **"NUDT2 is an enzyme important for maintaining the intracellular level of the diadenosine tetraphosphate (Ap4A),"** with compound-heterozygous variants (*p.R12*, *p.I65R*) *causing intellectual disability* ([PMID: 38243213](#)). *An earlier large ID cohort study also flagged NUDT2 among candidate genes harboring recessive variants* ([PMID: 27431290](#)). *Together these establish NUDT2 as a dual-function enzyme (Ap4A hydrolase + mRNA decapping)* that is essential for normal human neurodevelopment.*

F009 — Mouse Nudt2 sequence contains a canonical Nudix catalytic box (bioinformatic confirmation)

The mouse Nudt2 protein (P56380) is **147 aa (~17 kDa) with no transmembrane segments**. Motif scanning identifies the **Nudix signature** G-X₅-E-X₇-RE...EE...G ("GHVDPGENDLETALRETREETG," beginning at residue ~44), whose core "RETREETG" (res ~57–64) provides the conserved glutamates that coordinate the catalytic divalent metal (Mg²⁺/Zn²⁺). This aligns with the human ortholog's experimentally validated catalytic glutamate **E58**, probed by the E58A mutant whose crystal structure was solved ("**crystal structure of wild-type and E58A mutant human Ap4A hydrolase**," [PMID: 23384440](#)). The presence of the intact Nudix box in the mouse sequence is strong bioinformatic confirmation that the mouse enzyme is catalytically competent and mechanistically identical to the characterized human enzyme.

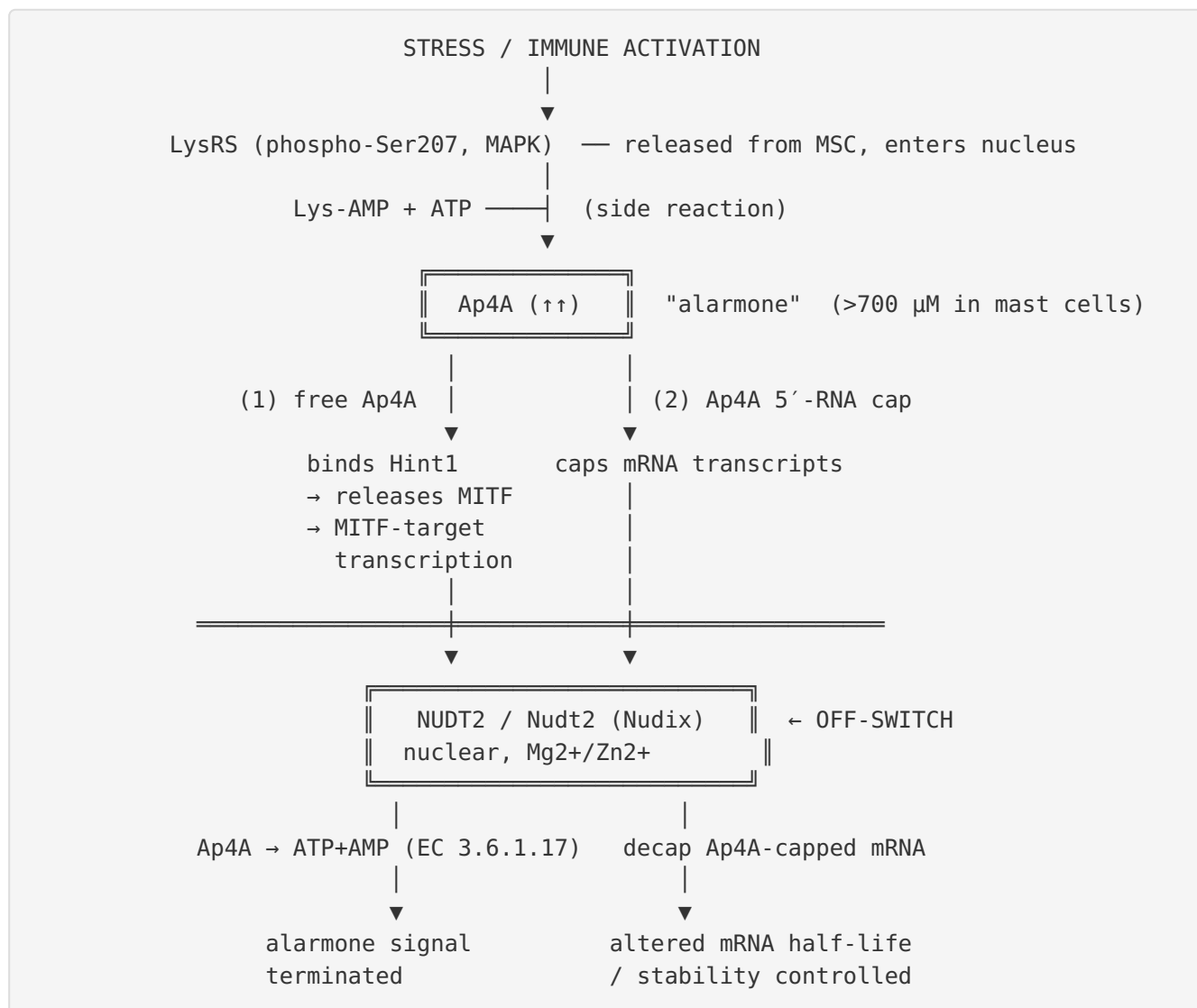
F010 — Mouse Nudt2 is highly conserved across mammals (89% identical to human), validating functional transfer

A Needleman–Wunsch global alignment shows mouse Nudt2 (P56380, 147 aa) is **89.1% identical to human NUDT2** (P50583) and **84.5% identical to pig NUDT2** (P50584) — all three annotated as "bis(5'-nucleosyl)-tetraphosphatase [asymmetrical]." The catalytic Nudix loop is essentially invariant (mouse ...RETREETGIE... vs pig ...RETQEEAGID...), preserving the metal-coordinating glutamates. UniProt assigns the mouse EC 3.6.1.17 **by similarity to the pig enzyme** (ECO:0000250|UniProtKB:P50584). The human ortholog's αβα-sandwich fold ([PMID: 23384440](#)) therefore provides a structural and mechanistic template that transfers directly to mouse Nudt2.

Ortholog	UniProt	% identity to mouse	Annotation
Mouse Nudt2	P56380	100% (reference)	Bis(5'-nucleosyl)-tetraphosphatase [asymmetrical]
Human NUDT2	P50583	89.1%	Bis(5'-nucleosyl)-tetraphosphatase [asymmetrical]
Pig NUDT2	P50584	84.5%	Bis(5'-nucleosyl)-tetraphosphatase [asymmetrical] (EC source)

Mechanistic Model / Interpretation

The findings converge on a unified model in which Nudt2 is the **degradative controller of the Ap4A alarmone** and a **regulator of mRNA stability via decapping**. Both activities share the same chemistry — Nudix-catalyzed, metal-dependent cleavage of a pyrophosphate bond within an adenosine-polyphosphate moiety — applied to two substrate contexts: free Ap4A, and Ap4A positioned as a 5' RNA cap.



Two linked but genetically separable activities. The add-back experiments in patient-derived cells ([PMID: 38141063](#)) are the linchpin: a NUDT2 variant that retains Ap4A hydrolysis but is defective in mRNA decapping fails to rescue mRNA homeostasis. This proves the two functions are distinct and that **decapping** is the activity controlling transcript stability, while **Ap4A hydrolysis** controls the free-alarmone pool.

Why loss of function is harmful. When Nudt2/ApaH is absent, free Ap4A accumulates (175-fold in human NuKO cells; high basal levels in bacterial *apaH* mutants), driving transcriptional reprogramming (interferon/host-response genes) and, in bacteria, heat hypersensitivity. In humans, biallelic loss causes a neurodevelopmental disorder. The consistent theme across kingdoms is that **the hydrolase is the essential homeostatic brake** on a signal that is otherwise rapidly generated under stress. Consistent with this essentiality, the Ap4A hydrolase is required for viability in the malaria parasite *Plasmodium berghei* ([PMID: 30700216](#)).

Localization logic. The enzyme is small, soluble, and predominantly nuclear with chromatin association — exactly where it can both terminate nuclear Ap4A signaling (acting downstream of nuclear LysRS) and participate in nuclear mRNA cap metabolism. Extracellular Ap4A is handled by a distinct ecto-phosphodiesterase (PC-1/PDase-I), so Nudt2's domain of action is the **intracellular/nuclear** compartment.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
23384440	<i>Crystal structure of wild-type and mutant human Ap4A hydrolase</i>	Defines asymmetrical cleavage "at the fourth phosphate," the $\alpha\beta\alpha$ Nudix fold, and catalytic glutamate E58 — structural/mechanistic template for mouse (F001, F009, F010)
17344088	<i>Characterisation of a bis(5'-nucleosyl)-tetraphosphatase (asymmetrical) from Drosophila</i>	Kinetics (K_m 9 μ M, k_{cat} 43 s^{-1}), ATP+AMP products, metal dependence, fluoride/ MgF_3^- inhibition, and nuclear localization (F002, F005)
9880487	<i>ialA nudix hydrolase active on dinucleoside polyphosphates (Bartonella)</i>	Broad Ap_nA/Gp_nG specificity with Ap4A preference; NTP + NMP products (F002)
27144453	<i>NUDT2 disruption elevates Ap4A, down-regulates immune/cancer genes</i>	175-fold Ap4A rise; 6,288 DEGs — NUDT2 is the dominant Ap4A regulator (F003)
14975237	<i>LysRS and Ap4A as signaling regulators of MITF</i>	$Ap4A \rightarrow Hint1 \rightarrow MITF$ signaling axis; Ap4A as second messenger (F004)
19524539	<i>LysRS as key signaling molecule in immune response</i>	Regulated Ap4A production (Ser207 phospho, nuclear translocation) upstream of Nudt2 (F004)
22329685	<i>Side chain independent recognition of aminoacyl adenylates by Hint1</i>	Structural basis of Hint1 adenylate surveillance that constrains Ap4A levels (F004)
9784621	<i>Bovine adrenal dinucleotide hydrolysis is PDase-I</i>	Separates extracellular Ap4A degradation (PC-1/PDase-I) from the intracellular Nudix enzyme (F005)
37934413	<i>Diadenosine tetraphosphate (Ap4A) as RNA cap</i>	Establishes Ap4A as a non-canonical 5' RNA cap — substrate class for decapping (F006)
1935909	<i>AppppA binds DnaK/GroEL; apaH mutants heat-sensitive</i>	Loss of Ap4A hydrolase $\rightarrow$ heat hypersensitivity; protective homeostatic role (F007)
6369319	<i>AppppA, heat-shock stress, and cell oxidation</i>	Original alarmone concept for Ap4A (F007)
38141063		Dual activity; decapping (not Ap4A hydrolysis) controls mRNA stability; human disease (F008)

PMID	Title (abbrev.)	How it supports the findings
	<i>Biallelic NUDT2 variants defective in mRNA decapping cause neurodevelopmental disease</i>	
38243213	<i>First ID case from compound-heterozygous NUDT2 variants</i>	Independent confirmation of NUDT2's Ap4A role and neurodevelopmental requirement (F008)
27431290	<i>Clinical genomics expands the morbid genome of ID</i>	Early identification of <i>NUDT2</i> as an ID candidate gene (F008)

Supporting context (Nudix "house-cleaning" family): Several papers frame Nudix hydrolases, including Ap4A hydrolases, as "house-cleaning" enzymes that sanitize nucleotide pools ([PMID: 16359314](#), [PMID: 15740738](#)), and the structural logic of Nudix substrate selection is exemplified by NUDT16 ([PMID: 26121039](#)). These reinforce the enzyme's placement and mechanistic class without altering the core functional assignment.

Challenges/nuances: No paper contradicts the core enzymatic assignment. The main nuance is that much of the precise biochemistry (kinetics, localization) comes from *orthologs* (*Drosophila*, human, pig, bacteria) rather than from the mouse protein directly — justified by the 89% identity and invariant catalytic box (F009, F010), but formally an inference.

Limitations and Knowledge Gaps

- 1. Direct mouse biochemistry is sparse.** The quantitative kinetics (Km, kcat, metal dependence) and subcellular localization data derive from orthologs (*Drosophila*, human, pig, bacterial). The mouse-specific assignment of EC 3.6.1.17 is "by similarity." While conservation is very high, a direct enzymatic characterization of recombinant mouse Nudt2 was not located.
- 2. Decapping activity not yet demonstrated for the mouse protein.** The mRNA-decapping function and its link to mRNA stability are established for human NUDT2 ([PMID: 38141063](#)); mouse-specific decapping assays and the physiological scope of Ap4A-capped transcripts in mouse tissues remain to be defined.
- 3. In vivo mouse phenotype unknown.** No *Nudt2* knockout mouse phenotype was identified in the reviewed literature. Whether mouse loss-of-function recapitulates the human neurodevelopmental phenotype is untested here.

4. **Localization resolution.** Nuclear localization is inferred from the *Drosophila* ortholog and the protein's physicochemical properties; a definitive mouse localization (nuclear vs. nucleocytoplasmic, chromatin association) has not been directly measured.
 5. **Signaling specificity.** The LysRS–Ap4A–Hint1–MITF axis is best characterized in mast cells/immune activation; the extent to which Nudt2 shapes this axis in neurons (the disease-relevant cell type) is not resolved.
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Proposed Follow-up Experiments / Actions

1. **Recombinant mouse Nudt2 enzymology.** Express and purify mouse Nudt2; measure Km/kcat for Ap4A, Ap5A, Ap6A, Gp4G; determine metal preference (Mg^{2+} vs Zn^{2+}) and fluoride sensitivity to confirm the MgF_3^- transition-state mechanism directly in the mouse enzyme.
 2. **Mouse mRNA-decapping assay.** Test whether mouse Nudt2 removes Ap4A 5' caps from synthetic Ap4A-capped RNA, and engineer a decapping-dead/hydrolysis-intact separation-of-function mutant (analogous to the human add-back) to dissect the two activities in mouse cells.
 3. **Localization.** Perform endogenous immunofluorescence / GFP-fusion imaging and subcellular fractionation in mouse cells to confirm nuclear/chromatin association predicted from the *Drosophila* ortholog.
 4. **Mouse genetic model.** Generate a *Nudt2* knockout or patient-mimic knock-in mouse; quantify intracellular Ap4A (expect a large increase), transcriptome changes (interferon/host-response signature), and neurodevelopmental/behavioral phenotypes to test translational relevance of the human disorder.
 5. **Neuronal pathway mapping.** Measure Ap4A dynamics and LysRS–Hint1–MITF activity in mouse neurons under stress, with and without Nudt2, to determine whether the alarmone axis operates in the disease-relevant cell type.
 6. **Ap4A-cap transcriptomics.** Apply Ap4A-cap-sequencing to mouse tissues $\pm$ Nudt2 to define the endogenous repertoire of Ap4A-capped transcripts and quantify how decapping shapes their half-lives.
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Conclusion

Mouse **Nudt2** (P56380, Apah1) is a small, soluble, predominantly nuclear **Nudix-family enzyme** whose primary function is the **metal-dependent asymmetrical hydrolysis of diadenosine tetraphosphate (Ap4A) and related dinucleoside polyphosphates to a nucleoside triphosphate plus a nucleoside monophosphate** ($\text{Ap4A} \rightarrow \text{ATP} + \text{AMP}$; EC 3.6.1.17). It is the **homeostatic off-switch for the Ap4A stress alarmone**, acting downstream of LysRS in the Ap4A–Hint1–MITF signaling axis, and it possesses a second, genetically separable **mRNA-decapping activity** that removes non-canonical Ap4A RNA caps and governs mRNA stability. Loss of this enzyme elevates Ap4A, reprograms interferon/host-response gene expression, and — in humans — causes a recessive neurodevelopmental disorder. The mouse protein's 89% identity to human NUDT2 and its intact, invariant Nudix catalytic box make these conclusions directly transferable to the mouse ortholog.

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